

Instructions

A meticulous gardener is interested in the length of blades of grass on his lawn. He believes that blade length X follows a normal distribution centered on 10 mm with a variance of 2 mm.

Write an R script which performs the following tasks:

1. Find the probability that a blade of grass is between 9.5 and 11 mm long.
Hint: "between" means subtract the cumulative probability at the lower bound from that at the upper bound.
2. What are the standardized values (i.e., Z scores) of 9.5 and 11 in the context of this distribution? Using the standardized values, confirm that you obtain the same probability you found in (1) with the standard normal density (in other words, repeat your calculation from (1) using Z scores).
Hint: Use the formula $Z = (X - \text{mean_value}) / \text{sd_value}$ in R to calculate the Z scores, where `sd_value` is the standard deviation, which is the square root of the variance. Then, use `pnorm()` to compute the probabilities for the Z scores and subtract the lower from the upper.
3. Below which value are the shortest 2.5 percent of blade lengths found?
Hint: For this percentage-based question, use the quantile function to find the blade length corresponding to the 2.5th percentile.
4. Standardize your answer from (3).
Hint: Use the formula $Z = (X - \text{mean_value}) / \text{sd_value}$ in R to convert the blade length to a Z score.

Non-Functional Requirements:

- Include a 4-line ID header at the beginning of your script.
- Include vertical spacing (a blank line) between logical blocks for readability.
- Comment your code thoughtfully (avoid excessive commenting).
- Ensure each line of code does not exceed 80 column

Expected Output:

```
Probability blade is between 9.5 and 11 mm: 0.3984131
Probability from Z-scores for 9.5 and 11 mm: 0.3984131
Blade length below shortest 2.5%: 7.228192 mm
Z-score of  shortest 2.5% blade: -1.959964
```